



PROBLEM STATEMENT

MScFE Capstone Project

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Project Track

Practical Track - Machine Learning (Deep) Investment Strategies: Quantitative Credit Risk Modeling

Problem statement

Traditional credit risk models, including Logistic Regression and standard credit scoring frameworks, assume that the relationship between borrower characteristics and probability of default (PD) is linear and stable over time (Bello 119).

However, empirical studies have shown that risk and particularly credit risk cannot be modeled linearly but rather non-linearly as a function of multiple variables, including internal borrower data, borrower behavior, macro and micro-economic conditions. Further, historical data from events like the 2008 financial crisis and the recent COVID-19 pandemic indicate that credit risk is highly non-linear and sensitive to macro and microeconomic shifts (Bello 110; Suhadolnik et al. 2, 20). Models calibrated during expansionary, low-volatility periods will systematically underestimate risk during downturns.

Drawing an analogy from asset pricing, the Capital Asset Pricing Model (CAPM) is extended by the Fama-French three-factor model by incorporating additional systematic risk factors such as size, value beyond market beta. Similarly, this project proposes to decompose credit risk into:

- Idiosyncratic borrower risk: captured by traditional credit features (leverage, interest coverage, liquidity, ...)
- Systematic macro-regime risk: driven by the macroeconomic environment and market regime at the time of assessment

This project aims to develop a macro-conditional and regime-dependent credit risk framework, where the probability of default is explicitly modeled as:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t, R_t)$$

- X_{borrower} : borrower-specific characteristics
- M_t : macroeconomic variables at time t
- R_t : latent market regime at time t

The objective is to determine whether incorporating macroeconomic conditions and latent regimes leads to statistically and economically significant improvements in credit risk modeling.

Goals and Objectives

The goal is to determine if adding latent regime shift dynamics and macroeconomic variables using advanced machine learning considerably enhances the resilience, calibration, and prediction accuracy of probability of default (PD) models across several economic cycles.

Hypothesis: ***“Macroeconomic and market regime information significantly improves predictive accuracy of PD models”***

Objective 1: Data Pipeline

- Load and align borrower-level loan data (Lending Club dataset) with macroeconomic time series and adjust frequency to monthly.
- Microeconomic data: Federal Funds Effective Rates (FEDFUNDS), Unemployment rate (UNRATE), Consumer Price Index for All Urban Consumers: All Items (CPIAUCSL), ICE BofA US High Yield Index Option-Adjusted Spread (BAMLH0A0HYM2), and Real Gross Domestic Product (GDPC1).
- Source: Federal Reserve Bank of St. Louis (FREDAPI)
- Feature engineering: Calculation of financial ratios (leverage, interest coverage ratio), cash flow volatility, and macroeconomic time series transformations

Objective 2: Credit Risk Modeling (Base Models)

- Implement and evaluate benchmark models: Logistic Regression, Gradient Boosting (LightGBM / XGBoost)
- Input: borrower-level features only
- Estimate probability of default:

$$\text{Probability of default } PD_{\text{baseline}} = Pr(\text{default} | X_{\text{borrower}})$$

Objective 3: Macroeconomic-Conditional Extension

- Incorporate macroeconomic variables: interest rates, inflation, unemployment, credit spreads, GDP growth proxies, ...
- Extended model:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t)$$

- Evaluate the performance between the extended models and the base models.

Objective 4: Regime Detection Layer

- Identify latent market regimes using unsupervised machine learning methods:
 - Hidden Markov Models (HMM)
 - Probabilistic clustering: Gaussian Mixture Models
 - Hierarchical clustering
 - Exclusive clustering: K-Means clustering
- Define interpretable regimes:
 - Volatility: low stress vs high stress
 - Growth: expansion vs recession
 - Liquidity: low vs high
- Assign regime label R_t to each observation

Objective 5: Regime-Dependent Credit Model

- Estimate regime-aware models:

$$PD = Pr(\text{default} | X_{\text{borrower}}, M_t, R_t)$$

- Compare:
 - Baseline vs macro vs regime-aware models
 - Stability across economic regimes

Objective 6: Model Evaluation

Evaluate all three model tiers using a structured framework:

Dimension	Metric	Purpose
Discrimination	ROC-AUC / PR-AUC	Ranking ability across borrower risk levels
Calibration	Brier Score	Probabilistic accuracy of PD estimates
Calibration	Calibration curves	Visual reliability of predicted vs. observed default rates
Robustness	Per-regime AUC	Stability of model performance across macro states
Robustness	Time-split OOS test	Out-of-sample generalization across economic cycles
Stress	2008-shock PD response	PD surface sensitivity under crisis-like macro conditions

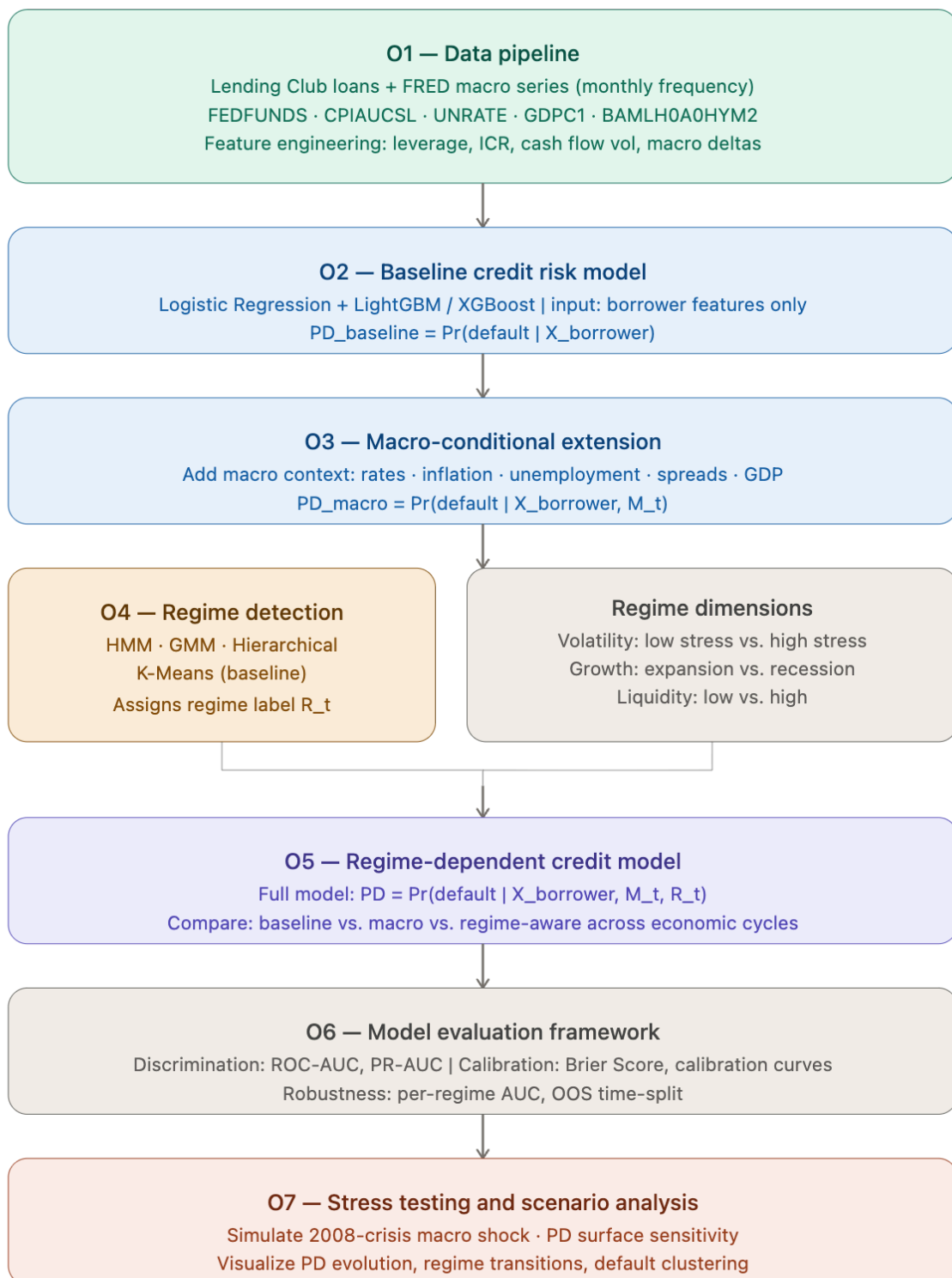
Objective 7: Stress Testing and Scenario Analysis

- Simulate adverse macroeconomic scenarios (for example, 2008 financial crisis macro conditions)
- Analyze PD sensitivity to macro shocks and default clustering effects
- Visualize PD surface evolution across macro states and regime transitions

Libraries and Data Sources

Category	Details
Data manipulation	pandas, numpy
Macro data access	fredapi (FRED API) or manual CSV download
Baseline modeling	scikit-learn, xgboost, lightgbm
Regime detection	hmmlearn, scikit-learn (GMM, KMeans, Agglomerative)
Evaluation & calibration	scikit-learn (metrics, calibration)
Visualization	matplotlib, seaborn, plotly
Loan dataset	Lending Club public dataset: Kaggle (2007–2018, labelled defaults)
Macro dataset	FRED (Federal Reserve Bank of St. Louis): free public API

Code Design Plan



References

Bello, Oluwabusayo Adijat. "Machine learning algorithms for credit risk assessment: an economic and financial analysis." *International Journal of Management* 10.1 (2023): 109-133.

Suhadolnik, Nicolas, Jo Ueyama, and Sergio Da Silva. "Machine learning for enhanced credit risk assessment: An empirical approach." *Journal of Risk and Financial Management* 16.12 (2023): 496.

Eugene F. Fama; Kenneth R. French. "The Capital Asset Pricing Model: Theory and Evidence".
<https://mba.tuck.dartmouth.edu/bespeneckbo/default/AFA611-Eckbo%20web%20site/AFA611-S6B-FamaFrench-CAPM-JEP04.pdf>

CFI. "Fama-French Three-Factor Model". Jan 28, 2000

https://corporatefinanceinstitute.com/resources/valuation/fama-french-three-factor-model/?utm_source=&utm_medium=cpc&utm_campaign=PMax_US&utm_term=&utm_content=&utm_matchtype=&utm_device=c&utm_ad=&cfi_gad_clid=CjwKCAjwhLPOBhBiEiwA8_wJHJM34nONCKv8Rcg9rDk1du03zLOSTW3kPUVLMzw1UqjU-jnigu4DjhoCvVsQAvD_BwE&campaign=PMax_US&adgroupid=&keyword=&device=c&network=x&placement=&adposition=&loc_physical_ms=9196251&loc_interest_ms=&campaignid=23700536377&gad_source=1&gad_campaignid=23700541429&gbraid=0AAAAAoJkld7X2Kb3-OLG7PsFFmDHdafLv&gclid=CjwKCAjwhLPOBhBiEiwA8_wJHJM34nONCKv8Rcg9rDk1du03zLOSTW3kPUVLMzw1UqjU-jnigu4DjhoCvVsQAvD_BwE